Linear Algebra

MATH10015 · Group 07 · The fields F_2 and F_3, vector spaces over them, and isomorphisms

FRIDAY 13 NOVEMBER 2026 · 09:00–10:00 · FRY BLDG G.14 (SAME ROOM AS WEEK 7 — CHECK TIMETABLE)

Linear Algebra — lesson at a glance

The fields F_2 and F_3, vector spaces over them, and isomorphisms.

Clock	What	Budget
09:00–09:05	Recall — 3 quick questions on last week, no notes	5 min
09:05–09:20	Round 1 — Arithmetic in F_2 and F_3 Watch: Tutor guide Week 8: warm-up and "new examples" · Try: Tutor guide Week 8: tables for F_3	15 min
09:20–09:35	Round 2 — Polynomials over F_2: counting, bases, and a surprise Watch: Tutor guide Week 8: "What do we mean by $P_n(K)$?" · Try: Tutor guide Week 8 (elements and bases of $P_n(F_2)$), with n = 2	15 min
09:35–09:48	Round 3 — Isomorphism: when are two spaces "the same"? **(optional)** Watch: Linear PS7 Q5(a)(b) · Try: Tutor guide Week 8: "decide if any of them are isomorphic"	13 min
09:48–09:55	Exit ticket — 3 questions alone, answers, "which round was hardest?", reminders	7 min

Cut rule: Rounds 1 and 2 are the lesson. If Round 2 is not wrapped up by **09:36**, skip Round 3 completely, keep working on Round 2, and start the exit ticket at 09:48. Round 3 then becomes the first item of "At home".

Opening words: "This week the scalars changed: instead of real numbers we allowed a field with only two or three elements. Everything from the last eight weeks still works — but some answers change. Three short rounds today: I show one, you do one, we compare."

> 开场 30 秒说玩法（同周四）。再说一句：交作业的是 PS7 第 1 题（下周三中午截止），今天不讲它。然后直接发 Recall。

BEFORE YOU START · tutor only

Deadlines and what has been lectured

- **Hand-in:** Problem Sheet 7, **Question 1** (a basis of $M_2(\mathbb{R})$), due **noon on Wednesday 18 November** (Week 9). Pointers only — do not work it out in the room.
- Problem Sheet 6 Q1 closed on Wednesday 11 November; feedback appears in the Gradebook tab.
- There is a Numbas quiz with Sheet 7, and the usual Friday drop-in (3–5pm, G.10 — as announced on 5 November).

> **本周内容来自 Rachael 11 月 10 日发的 tutor guide 第 8 周**（上周日的草稿是没等到 guide 时做的"第 7 章复习备用版"，已换掉）。guide 的活动：解释 $\mathbb{F}_2$、写 $\mathbb{F}_2/\mathbb{F}_3$ 的运算表、列 $(\mathbb{F}_2)^2$ 的元素、$P_n(K)$ 的元素和两组基、判断哪些空间同构、PS7 Q5 和 Q7。今天三轮覆盖前六项；Q7 留作课后（它和交作业的 Q1 都在 $M_2(\mathbb{R})$ 上，课上讲 Q7 容易顺口把 Q1 讲出来）。guide 还提醒：本周内容多，学生可能会问具体哪个对象看不懂——Try 的时候巡视正好听这些问题。

TUTOR · HOW TO READ THE SYMBOLS ALOUD

Symbol reading table / 符号读法

Symbol	Read aloud	中文
$\mathbb{F}_2$	“F two”, the field with two elements	只有 0、1 两个元素的域
$\mathbb{F}_3$	“F three”	0、1、2，模 3 运算
$(\mathbb{F}_2)^n$	“F two to the n”	分量取自 $\mathbb{F}_2$ 的 n 维向量
$P_n(K)$	“P n of K”: polynomials of degree at most n over K	次数 ≤ n、系数在 K 中的多项式
$M_2(\mathbb{R})$	“M two of R”: 2 by 2 real matrices	2×2 实矩阵
$V \cong W$	“V is isomorphic to W”	同构
$\dim_K V$	“the dimension of V over K”	V 作为 K 上向量空间的维数
φ	“phi”	
$\mathrm{Span}(v_1, v_2)$	“the span of v one, v two”	张成

PRINTED AT THE TOP OF THE HANDOUT

Key ideas (students have this box)

1. **Small fields.** $\mathbb{F}_2 = \{0, 1\}$ with arithmetic mod 2 (so $1 + 1 = 0$); $\mathbb{F}_3 = \{0, 1, 2\}$ with arithmetic mod 3. Every non-zero element has a multiplicative inverse.
2. **Vectors over $\mathbb{F}_p$.** $(\mathbb{F}_p)^n$ has entries *and scalars* from $\mathbb{F}_p$, so it contains exactly p^n vectors.
3. **Polynomial spaces.** $P_n(K)$ = polynomials of degree at most n with coefficients in K; basis $1, x, \ldots, x^n$; dimension $n + 1$.
4. **Isomorphism.** A linear map that is bijective. If $\dim_K V = n$ then $V \cong K^n$; over the same field, same (finite) dimension $\iff$ isomorphic.

RECALL · 09:00–09:05 · 5 min

Recall: three quick questions

No notes, 2 minutes on your own, then compare with your neighbour.

1. Write down the two conditions for a map $T : V \to W$ to be linear.

   $T(u + v) = T(u) + T(v)$ and $T(\lambda v) = \lambda T(v)$ for all $u, v \in V$ and all scalars λ.

2. $T : \mathbb{R}^3 \to \mathbb{R}^2$ is linear and surjective. What is its nullity?

   Rank $= 2$, so by Rank–Nullity the nullity is $3 - 2 = 1$.

3. True or false: if T is linear and $\ker T = \{0\}$, then T is injective.

   True: $T(u) = T(v) \Rightarrow T(u - v) = 0 \Rightarrow u - v \in \ker T = \{0\} \Rightarrow u = v$.

> 2 分钟自己写、1 分钟同桌对、2 分钟你报答案。三题都是上周第 7 章的，第 3 轮（同构 = 线性 + 双射）全要用到。不展开讲。

ROUND 1 · 09:05–09:20 · 15 min

Arithmetic in F_2 and F_3

Skill: Add, multiply and invert in a small field; count the vectors in $(F_p)^n$.

WATCH · 09:05–09:09 · you at the board

Tutor guide Week 8: warm-up and "new examples"

(i) Write the addition and multiplication tables of $\mathbb{F}_2$.

(ii) List the elements of $(\mathbb{F}_2)^2$. What is $v + v$ for a vector v in this space?

> **中文题意：**(i) 写出 $\mathbb{F}_2$ 的加法表和乘法表；(ii) 列出 $(\mathbb{F}_2)^2$ 的全部 4 个向量，并说明 $v + v = 0$。

> **Say:** Think of a light switch: 0 is off, 1 is on, and adding 1 flips it. Flip twice and you are back where you started — that is one plus one equals zero.

$+$	**0**	**1**
0	0	1
1	1	0

$\times$	**0**	**1**
0	0	0
1	0	1

Only one surprise: $1 + 1 = 0$. So $-1 = 1$ in $\mathbb{F}_2$.

$(\mathbb{F}_2)^2 = \{(0,0),\ (1,0),\ (0,1),\ (1,1)\}$: two choices for each entry, so $2^2 = 4$ vectors. The scalars are only 0 and 1.

For any v: $v + v = (1 + 1)v = 0 \cdot v = 0$. Example: $(1,1) + (1,1) = (0,0)$, and $(1,0) + (1,1) = (0,1)$.

> 4 分钟。guide 的热身是"你会怎么向两周前的自己解释 $\mathbb{F}_2$？"——可以先问一句、听一个回答，再给出"开关"说法。两张表画在黑板上并留着，学生做 $\mathbb{F}_3$ 时照着格式画。

TRY · 09:09–09:15 · pairs, you walk round

Tutor guide Week 8: tables for F_3 · on the handout

(a) Fill in the addition and multiplication tables of $\mathbb{F}_3 = \{0, 1, 2\}$.

(b) In $\mathbb{F}_3$: what is -1? What is 2^{-1}?

(c) Solve $2x + 1 = 0$ in $\mathbb{F}_3$.

(d) How many vectors are there in $(\mathbb{F}_3)^2$?

$+$	**0**	**1**	**2**
0			
1			
2			

$\times$	**0**	**1**	**2**
0			
1			
2			

> **中文题意：**(a) 填 $\mathbb{F}_3$ 的两张运算表（讲义上印了空表）；(b) -1 和 2^{-1} 各是多少；(c) 在 $\mathbb{F}_3$ 里解 $2x + 1 = 0$；(d) $(\mathbb{F}_3)^2$ 有多少个向量。

While you walk round — say this, not the answer:

- Wrote $2^{-1} = 0.5$: “which of 0, 1, 2 gives 1 when you multiply it by 2?”
- “-1 is not in the set”: “which element do you add to 1 to get 0?”
- For (d) answered 6 or 4: “how many choices for the first entry? the second?”

Fast finishers (printed on the handout): Finished early? Is $\{0, 1, 2, 3\}$ with arithmetic mod 4 a field? (Look at 2×2.)

$+$	**0**	**1**	**2**
0	0	1	2
1	1	2	0
2	2	0	1

$\times$	**0**	**1**	**2**
0	0	0	0
1	0	1	2
2	0	2	1

(b) $-1 = 2$ because $1 + 2 = 0$. $2^{-1} = 2$ because $2 \times 2 = 4 = 1$.

(c) $2x + 1 = 0 \iff 2x = -1 = 2 \iff x = 1$ (multiply by $2^{-1} = 2$). Check: $2 \cdot 1 + 1 = 3 = 0$.

(d) $3 \times 3 = 9$.

Fast finishers: no — mod 4 we get $2 \times 2 = 0$, so 2 has no multiplicative inverse. (This is why we only use a prime p.)

EXPLAIN · 09:15–09:20 · a pair reports

How: Two pairs come up and fill in one table each; the class corrects. Then take (b)–(d) by voice.

> **The trap:** Fractions and minus signs still exist — but they are elements of the field: in $\mathbb{F}_3$, "$\frac{1}{2}$" is 2 and "-1" is 2.

> **Takeaway (students write it in their own words):** In $\mathbb{F}_p$: do ordinary arithmetic, then reduce mod p. $(\mathbb{F}_p)^n$ has p^n vectors.

> **Quick check:** How many vectors are in $(\mathbb{F}_2)^3$? And in $(\mathbb{F}_3)^3$?
>
> $2^3 = 8$ and $3^3 = 27$.

5 分钟。让两组上黑板各填一张表，其他人挑错，比你自己写更省时间。重点只有一句："分数和负号还在，只是它们就是 0、1、2 里的某一个。"

ROUND 2 · 09:20–09:35 · 15 min

Polynomials over F_2: counting, bases, and a surprise

Skill: List a vector space over F_2, find bases, test linear independence over F_2.

WATCH · 09:20–09:24 · you at the board

Tutor guide Week 8: "What do we mean by $P_n(K)$?"

$P_1(\mathbb{F}_2)$ is the set of polynomials $a_0 + a_1 x$ with $a_0, a_1 \in \mathbb{F}_2$.

(i) List its elements. (ii) Give a basis. (iii) Is $\{1,\ 1+x\}$ also a basis?

中文题意： $P_1(\mathbb{F}_2)$：系数取 0 或 1、次数不超过 1 的多项式。(i) 列出全部 4 个元素；(ii) 给一组基；(iii) $\{1, 1+x\}$ 是不是也是基（是）。

Say: A polynomial here is just a list of coefficients, and each coefficient is 0 or 1. So counting the polynomials is the same as counting the lists.

(i) Two choices for a_0, two for a_1: $P_1(\mathbb{F}_2) = \{0,\ 1,\ x,\ 1+x\}$, four elements.

(ii) $\{1, x\}$: every element is $a_0 \cdot 1 + a_1 \cdot x$ in exactly one way. So $\dim P_1(\mathbb{F}_2) = 2$.

(iii) Yes. *Independent:* $\lambda \cdot 1 + \mu(1+x) = (\lambda+\mu) + \mu x = 0$ forces $\mu = 0$, then $\lambda = 0$. *Spanning:* $x = 1 + (1+x)$, because $1 + 1 = 0$. Two independent vectors in a space of dimension 2 form a basis.

4 分钟。把 $P_1(\mathbb{F}_2)$ 的 4 个元素和 $(\mathbb{F}_2)^2$ 的 4 个向量并排写：$a_0 + a_1 x \leftrightarrow (a_0, a_1)$——这是给第 3 轮"同构"埋的伏笔，不用点破。

TRY · 09:24–09:30 · pairs, you walk round

Tutor guide Week 8 (elements and bases of $P_n(F_2)$), with n = 2 · on the handout

Now take $P_2(\mathbb{F}_2)$: polynomials $a_0 + a_1x + a_2x^2$ with coefficients in $\mathbb{F}_2$.

(a) How many elements does it have? List them.

(b) Write down two *different* bases.

(c) Is $\{\,1+x,\ \ x+x^2,\ \ 1+x^2\,\}$ linearly independent over $\mathbb{F}_2$? (Try adding all three.)

> **中文题意：** $P_2(\mathbb{F}_2)$：(a) 有几个元素并列出来（8 个）；(b) 写两组不同的基；(c) $\{1+x,\ x+x^2,\ 1+x^2\}$ 在 $\mathbb{F}_2$ 上线性无关吗（相关——三个加起来是 0；在 $\mathbb{R}$ 上却是无关的）。guide 原文用的是 $P_3(\mathbb{F}_2)$（16 个元素），这里先用 P_2 好列举，P_3 放在“提前做完”里。

While you walk round — say this, not the answer:

- Listing slowly: “how many choices for each coefficient? so how many polynomials?”
- Second basis: “swap one basis vector for a sum of two of them — is it still a basis?”
- For (c) set up three equations over $\mathbb{R}$: “fine — now, what is 2 in $\mathbb{F}_2$?”

Fast finishers (printed on the handout): Finished early? How many elements does $P_3(\mathbb{F}_2)$ have? Which elements of $P_2(\mathbb{F}_2)$ are *not* in $\mathrm{Span}(1+x,\ x+x^2)$?

(a) $2^3 = 8$ elements:

$$0,\ \ 1,\ \ x,\ \ 1+x,\ \ x^2,\ \ 1+x^2,\ \ x+x^2,\ \ 1+x+x^2.$$

(b) For example $\{1,\ x,\ x^2\}$ and $\{1,\ 1+x,\ 1+x+x^2\}$. (For the second: $x = 1 + (1+x)$ and $x^2 = (1+x) + (1+x+x^2)$, so it spans; three spanning vectors in dimension 3 form a basis.) Many other answers are possible — every basis has exactly 3 elements.

(c) No:

$$(1+x) + (x+x^2) + (1+x^2) = 2 + 2x + 2x^2 = 0 \quad \text{in } \mathbb{F}_2,$$

a linear combination with coefficients $1, 1, 1$ (not all zero) that gives 0. So the set is linearly **dependent** over $\mathbb{F}_2$.

Over $\mathbb{R}$ the same three polynomials are independent: $\lambda(1+x) + \mu(x+x^2) + \nu(1+x^2) = 0$ gives $\lambda + \nu = \lambda + \mu = \mu + \nu = 0$, hence $2\nu = 0$, so $\nu = 0$ — but in $\mathbb{F}_2$, $2\nu = 0$ is true for every ν.

Fast finishers: $P_3(\mathbb{F}_2)$ has $2^4 = 16$ elements. $\mathrm{Span}(1+x,\ x+x^2) = \{0,\ 1+x,\ x+x^2,\ 1+x^2\}$, so the other four — $1,\ x,\ x^2,\ 1+x+x^2$ — are not in it.

EXPLAIN · 09:30–09:35 · a pair reports

How: Collect the 8 elements from the room in order; ask two pairs for their second basis and check one of them together; then vote on (c) before showing the sum.

> **The trap:** Linear independence depends on the field. The same three polynomials are independent over $\mathbb{R}$ and dependent over $\mathbb{F}_2$.

> **Takeaway (students write it in their own words):** A space of dimension n over $\mathbb{F}_p$ has exactly p^n vectors. Always ask: independent *over which field*?

> **Quick check:** A vector space over $\mathbb{F}_3$ has 81 vectors. What is its dimension?
>
> $81 = 3^4$, so the dimension is 4.

5 分钟。(c) 先让全班举手表决“无关 / 相关”，再把三个加起来——多数人会猜无关，这个反差就是这一轮要他们带走的东西。到 09:36 还没收完就跳过第 3 轮。

ROUND 3 · 09:35–09:48 · 13 min · OPTIONAL — skip if Round 2 runs past 09:36

Isomorphism: when are two spaces “the same”?

Skill: Show a map is an isomorphism; decide by dimension which spaces are isomorphic.

WATCH · 09:35–09:39 · you at the board

Linear PS7 Q5(a)(b)

Let $\varphi : \mathbb{R}^3 \to P_2$ send e_1 to 1, e_2 to x, and e_3 to x^2.

(a) If φ is linear, what is the output of a general $(a_1, a_2, a_3) \in \mathbb{R}^3$?

(b) Is φ an isomorphism? What do you need to prove?

中文题意： $\varphi : \mathbb{R}^3 \to P_2$ 把 e_1, e_2, e_3 送到 $1, x, x^2$。(a) 若线性，(a_1, a_2, a_3) 的像是什么（$a_1 + a_2x + a_3x^2$）；(b) φ 是同构吗、要证什么（线性 + 单射 + 满射；是）。

Say: An isomorphism is a dictionary between two spaces: it translates every vector, loses nothing, and respects addition and scaling. Here the dictionary reads a column of three numbers as the coefficients of a polynomial.

(a) $(a_1, a_2, a_3) = a_1e_1 + a_2e_2 + a_3e_3$, so by linearity

$$\varphi(a_1, a_2, a_3) = a_1\varphi(e_1) + a_2\varphi(e_2) + a_3\varphi(e_3) = a_1 + a_2x + a_3x^2.$$

(b) Isomorphism = linear *and* bijective. We are assuming linear, so check:

Injective: if $\varphi(a_1, a_2, a_3) = 0$ then $a_1 + a_2x + a_3x^2$ is the zero polynomial, so $a_1 = a_2 = a_3 = 0$. Hence $\ker \varphi = \{0\}$.

Surjective: any $p = c_0 + c_1x + c_2x^2 \in P_2$ equals $\varphi(c_0, c_1, c_2)$.

So φ is an isomorphism: $\mathbb{R}^3 \cong P_2$. *Shortcut:* nullity 0 gives rank $3 = \dim P_2$ by Rank–Nullity, so surjectivity is automatic.

4 分钟。黑板上写“isomorphism = linear + injective + surjective”三个词，逐个打勾。捷径（Rank–Nullity）正是 Recall 第 2、3 题。PS7 Q5(c)(d) 留作课后。

TRY · 09:39–09:45 · pairs, you walk round

Tutor guide Week 8: “decide if any of them are isomorphic” · on the handout

Fill in a basis and the dimension of each space *over the field shown*. Then decide: which of these spaces are isomorphic to each other?

Space	over	a basis	dim
$\mathbb{R}^2$	$\mathbb{R}$		
$\mathbb{C}^2$	$\mathbb{C}$		
$\mathbb{C}^2$	$\mathbb{R}$		
$P_2(\mathbb{R})$	$\mathbb{R}$		
$P_1(\mathbb{R})$	$\mathbb{R}$		

> **中文题意：** guide 给的五个空间：$\mathbb{R}^2$（$\mathbb{R}$ 上）、$\mathbb{C}^2$（$\mathbb{C}$ 上）、$\mathbb{C}^2$（$\mathbb{R}$ 上）、$P_2(\mathbb{R})$、$P_1(\mathbb{R})$。讲义上是一张表：填一组基和维数，再判断哪些互相同构。

While you walk round — say this, not the answer:

- $\mathbb{C}^2$ over $\mathbb{R}$ given dimension 2: “can you get $(i, 0)$ from $(1, 0)$ and $(0, 1)$ using only real scalars?”
- Says $\mathbb{C}^2$ over $\mathbb{C}$ is isomorphic to $\mathbb{R}^2$: “what would a linear map between them have to do with the scalar i?”

Fast finishers (printed on the handout): Finished early? Write down an explicit isomorphism between the two isomorphic spaces. Which $\mathbb{R}^n$ is $\mathbb{C}^2$ (over $\mathbb{R}$) isomorphic to?

Space	over	a basis	dim
$\mathbb{R}^2$	$\mathbb{R}$	e_1, e_2	2
$\mathbb{C}^2$	$\mathbb{C}$	e_1, e_2	2
$\mathbb{C}^2$	$\mathbb{R}$	$e_1,\ ie_1,\ e_2,\ ie_2$	4
$P_2(\mathbb{R})$	$\mathbb{R}$	$1, x, x^2$	3
$P_1(\mathbb{R})$	$\mathbb{R}$	$1, x$	2

Isomorphic: $\mathbb{R}^2 \cong P_1(\mathbb{R})$ — both have dimension 2 over $\mathbb{R}$; an isomorphism is $(a, b) \mapsto a + bx$.

Not isomorphic to anything else in the list: $\mathbb{C}^2$ over $\mathbb{R}$ (dimension 4, so it is $\cong \mathbb{R}^4$) and $P_2(\mathbb{R})$ (dimension 3, so $\cong \mathbb{R}^3$ — that was the Watch example).

$\mathbb{C}^2$ **over** $\mathbb{C}$**:** dimension 2, but over a different field. An isomorphism is a linear map, and “linear” only makes sense between spaces over the *same* field, so we do not compare it with $\mathbb{R}^2$. (Regarded over $\mathbb{R}$ it is the third row, of dimension 4.)

EXPLAIN · 09:45–09:48 · a pair reports

How: Ask for the dimension column by voice, then "which two are isomorphic?" and one sentence why.

> **The trap:** The dimension depends on the field: $\mathbb{C}^2$ has dimension 2 over $\mathbb{C}$ but 4 over $\mathbb{R}$.

> **Takeaway (students write it in their own words):** Isomorphism = linear + bijective. Over the same field: same dimension $\iff$ isomorphic. $\dim_K V = n \Rightarrow V \cong K^n$.

> **Quick check:** Is $M_2(\mathbb{R})$ isomorphic to $\mathbb{R}^4$? (Just yes or no and a reason — the map is PS7 Q7.)
>
> Yes: both have dimension 4 over $\mathbb{R}$.

3 分钟。这一轮可选：第 2 轮 09:36 前没收完就整轮跳过，Q5 留作课后。最后那个检查题只要"是，维数都是 4"就停——不要展开 $M_2(\mathbb{R})$ 的基，那是交作业的 Q1。

EXIT TICKET · 09:48–09:55 · 7 min

Exit ticket: three questions, on your own

1. In $\mathbb{F}_3$, compute $2+2$ and 2×2.

   $2+2=1$ and $2\times 2=1$.

2. How many vectors are there in $(\mathbb{F}_2)^3$?

   $2^3=8$.

3. Is $P_3(\mathbb{R})$ isomorphic to $\mathbb{R}^4$? Give a one-line reason.

   Yes: both have dimension 4 over $\mathbb{R}$ (basis $1, x, x^2, x^3$).

Then: "Circle the round you are least sure about — hands up for Round 1? Round 2? Round 3?" Note the count; next week's Recall comes from the round with most hands.

3 分钟各自写，2 分钟报答案、学生自己打勾，然后圈出"最没把握的一轮"并举手统计。最后提醒：PS7 第 1 题下周三中午交；Numbas quiz；周五 3–5 点 G.10 drop-in。跳过了第 3 轮的话，第 3 题改成口头问并顺带说一句"维数相同就同构"。

After the tutorial (printed on page 1 of the handout)

- **From today:** Problem Sheet 7 Q2 (subspaces of $(\mathbb{F}_2)^n$ — follows on from Round 1), Q5(c)(d) and Q7 (isomorphisms).
- **Then:** Q3, Q4, and the Numbas quiz.
- **Hand-in:** Q1, by noon on Wednesday 18 November.

Solutions to the additional questions are on Blackboard with Problem Sheet 7.

LOOKUP · hand-in pointers and the rest of today's Sheet 7 questions

Answer bank — Linear

Consider $B = \{A_1, A_2, A_3, A_4\} \subseteq M_2(\mathbb{R})$ with

$$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},\ A_2 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},\ A_3 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},\ A_4 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

(a) Informally explain why $E = \{E_{11}, E_{12}, E_{21}, E_{22}\}$ is a basis for $M_2(\mathbb{R})$ over $\mathbb{R}$. (b) Assuming E is a basis, determine whether B forms a basis. (c) Write the matrix with 1's in each coordinate as a linear combination in B.

Pointers only — do not decide (b) or compute (c) in the room.

- (a) What is E_{ij}? How do we add matrices and multiply them by scalars? Then: can every matrix be written using the E_{ij}, and in how many ways?
- (b) How many matrices are in B, and what is $\dim M_2(\mathbb{R})$? So which *one* of "spanning" and "linearly independent" is enough to check? Set up $\lambda_1 A_1 + \cdots + \lambda_4 A_4 = 0$ entry by entry.
- (c) Set up the same kind of system with the all-ones matrix on the right-hand side.
- Rachael's marking comments from previous sheets: introduce your notation, state what you are checking, write in sentences.

Let V be a vector space over $\mathbb{F}_2$ (so $v + v = 0$ for all $v \in V$). (a) Which part of the subspace test is unnecessary to check in this instance? (b) Hence determine the number of subspaces of $V = (\mathbb{F}_2)^n$ for $n = 1, 2, 3$. (c) A fellow student claims that the same simplification in (a) applies if V is a vector space over $\mathbb{F}_3$. Consider whether you agree with them.

(a) Closure under scalar multiplication. The only scalars are 0 and 1; $1 \cdot u = u$ and $0 \cdot u = 0$, which is in U as soon as U contains 0. So U is a subspace $\iff$ $0 \in U$ and U is closed under addition.

(b) $n = 1$: $\{0\}$ and V: **2**. $n = 2$: $\{0\}$; the three sets $\{0, v\}$ with $v \neq 0$; V: **5**. $n = 3$: $\{0\}$; seven sets $\{0, v\}$; seven subspaces of dimension 2 (each $\{0, u, v, u + v\}$; there are $7 \cdot 6/6 = 7$ of them); V: **16**.

(c) Agree. In $\mathbb{F}_3$ the scalars are $0, 1, 2$ and $2u = u + u$, so closure under addition (with $0 \in U$) already gives closure under scalar multiplication. The same works over any $\mathbb{F}_p$, since every scalar is $1 + 1 + \cdots + 1$.

Derived for this guide; compare with the official solutions on Blackboard before quoting.

(c) Imagine that a and b lie on the same line in $\mathbb{R}^3$, and that this line passes through 0. What can we deduce about $\varphi(a)$ and $\varphi(b)$? (d) For extra practice, work out the image of certain lines or planes under φ.

(c) If $a \neq 0$ then $b = \lambda a$ for some $\lambda \in \mathbb{R}$, so $\varphi(b) = \lambda \varphi(a)$: the two polynomials are scalar multiples of each other, i.e. they lie on the same line through 0 in P_2. (If $a = 0$ then $\varphi(a) = 0$.)

(d) Examples: the line $\mathrm{Span}(e_1 + e_2)$ goes to $\{t(1 + x) : t \in \mathbb{R}\}$; the plane $a_3 = 0$ goes to P_1, the polynomials of degree at most 1; the plane $a_1 = 0$ goes to the polynomials with $p(0) = 0$. An isomorphism sends subspaces to subspaces of the same dimension.

Let $\varphi : \mathbb{R}^4 \to M_2(\mathbb{R})$ be defined by $(w, x, y, z) \mapsto \begin{pmatrix} w & x \\ y & z \end{pmatrix}$. (a) Show that φ is linear. (b) Show that φ is injective. (c) Show that φ is surjective. (Is there a shortcut here?) (d) What have we shown in (a)–(c)?

(a) Matrices are added and scaled entry by entry, so $\varphi(u + v) = \varphi(u) + \varphi(v)$ and $\varphi(\lambda u) = \lambda \varphi(u)$: compare the four entries on each side.

(b) If $\varphi(u) = \varphi(v)$, the two matrices have the same four entries, so $u = v$. (Or: $\ker \varphi = \{0\}$, since only $(0, 0, 0, 0)$ goes to the zero matrix.)

(c) $\begin{pmatrix} p & q \\ r & s \end{pmatrix} = \varphi(p, q, r, s)$. *Shortcut:* nullity 0 gives rank $4 = \dim M_2(\mathbb{R})$ by Rank–Nullity.

(d) φ is an isomorphism, so $\mathbb{R}^4 \cong M_2(\mathbb{R})$.

Tutor note: the hand-in Q1 is also about $M_2(\mathbb{R})$. If you discuss Q7, stop at (d) and do not go on to bases of $M_2(\mathbb{R})$.

Sources and scope

Pure: Problem Sheet 7 with solutions (attached to Tim Burness's email "Intro to Pure: End of Week 7 updates", 6 Nov 2026); Blackboard "Lecture summary" read on 11 Nov 2026 (lectures of 9 and 10 Nov: Sections 7.4–7.5); announcement "Week 8 updates" (9 Nov 2026).

Linear: "Linear Algebra tutorial ideas 2026", Week 8 section (R. Carey, emailed 10 Nov 2026); Linear Algebra Problem Sheet 7 (Blackboard, read 11 Nov 2026); Charles Cox's announcement "Week 7 recap and coming up in week 8" (5 Nov 2026).

Hand-in questions are hints only: Pure PS7 Assigned Question (assessed, due 13:00 Thu 19 Nov), Pure PS6 Assigned Question (due 13:00 Thu 12 Nov), Linear PS7 Q1 (due noon Wed 18 Nov). The student handouts contain questions, key ideas and working space only — no answers.

> 本页给老师看（含答案），不要发给学生；学生只拿 Student PDF。